Sandeep Kushawah

Computational Soft Matter · Statistical Physics of the Glass Transition

sandeep.kushawah@outlook.com | ORCID: 0009-0003-5052-4398

sandeepkushawah.github.io | linkedin.com/in/sandeep-kushawah | Google Scholar

Research Interests

Computational soft matter and the statistical physics of the glass transition. My research uses large-scale molecular dynamics simulations to characterise structure–dynamics relationships in supercooled glass-forming liquids. Current work focuses on developing structural and orientational order parameters, configurational/structural entropy measures, and static and dynamic correlation lengths that connect local packing to slow, heterogeneous dynamics in model binary mixtures near the glass transition.

Education

Ph.D. in Physics (thesis submission: Oct 2026) — since Feb 2021

Indian Institute of Technology Mandi, India

Advisor: Dr. Prasanth P. Jose (Associate Professor)

- Thesis: *A Study of Structural and Dynamical Inhomogeneities in Supercooled Glass-Forming Model Liquids*; synopsis seminar presented 1 Oct 2026.
- Coursework CGPA: 8.0/10 (13 credits).

M.Sc. in Physics — July 2017 – May 2019

Indian Institute of Technology Delhi, India

- CGPA: 6.65/10 (80 credits).

B.Sc. (Hons.) in Physics — July 2014 – June 2017

Shivaji College, University of Delhi, India

- Minors in Mathematics and Computer Science. Aggregate: 71.33%.

Publications

Peer-reviewed journal articles

[1] Devansu Chakraborty, **Sandeep Kushawah**, Manoj Rana, Prasanth P. Jose. Breakdown of Stokes–Einstein–Debye relations at the gas–supercooled liquid coexistence. *Soft Matter*, 2026 (accepted).

[2] **Sandeep Kushawah**, Devansu Chakraborty, Prasanth P. Jose. Beyond pair entropy: orientational many-body correlations in supercooled glass-forming liquids from a four-point structural entropy. *Soft Matter*, 2026, **22**, 4360–4370. doi:10.1039/D6SM00491A. *(first & corresponding author)*

Conference proceedings

[3] **Sandeep Kushawah**, Shubhanjali Pathak, Prasanth P. Jose. Effect of minority composition in a binary glass forming liquid. *AIP Conf. Proc.*, 2025, **3198**(1), 020050. doi:10.1063/5.0248436.

[4] Keshav Thakur, **Sandeep Kushawah**, Prasanth P. Jose. Relation between local density Debye–Waller factor and density relaxation during glass transition in model systems. *AIP Conf. Proc.*, 2025, **3198**(1), 020051. doi:10.1063/5.0248437.

Research Experience

Doctoral Researcher, IIT Mandi — Feb 2021 – present

Physics of Complex Systems Lab, supervised by Dr. Prasanth P. Jose

- Simulate Kob–Andersen 80:20 binary Lennard-Jones glass-forming mixtures using large-scale molecular dynamics.
- Develop and analyse static structural/orientational order parameters and configurational entropy measures, and relate them to dynamical heterogeneity and structural relaxation.
- Implement and validate custom C/MATLAB analysis pipelines for radial and angular distribution functions and correlation-length extraction.

Master's Thesis – Irreversible Aggregation of Asymmetrical Patchy Particles — July 2018 – June 2019

Supervised by Dr. Sujin B. Babu (IIT Delhi)

- Simulated asymmetrical patchy colloids using Patchy Brownian Cluster Dynamics to study protein aggregation relevant to neurodegenerative disease.

- Visualised configurations using POV-Ray.

Master's Mini Project – Simulating Plasma using OOPIC Jan 2019 – June 2019
Supervised by Dr. Vikrant Saxena (IIT Delhi)

- Simulated plasma in a cylindrical chamber using a particle-in-cell model and characterised its behaviour.

Technical Skills

Simulation & modelling: Molecular Dynamics, Molecular Simulation, LAMMPS, GROMACS, VMD, POV-Ray.

Programming & analysis: C/C++, Python, MATLAB, Bash scripting, LaTeX.

Computing: High-Performance Computing (HPC), SLURM, PBS; Linux (CentOS, Ubuntu), Windows; MS Office.

Languages: Hindi (native), English (professional working proficiency).

Selected Presentations & Schools

- Poster, **COMPFLU 2025**, IISc Bangalore, Dec 2025 – "Temperature-dependent string-like cooperative motions in Kob–Andersen binary mixtures".
- **Computation and Modeling for Complex Fluids** pre-conference workshop (COMPFLU 2025), IISc Bangalore, Dec 2025 (selected participant).
- Poster, **68th DAE Solid State Physics Symposium**, Bhabha Atomic Research Centre, Mumbai, India.
- Poster, **67th DAE Solid State Physics Symposium**, Bhabha Atomic Research Centre, Mumbai, India.
- **Bangalore School on Statistical Physics XV**, ICTS & Raman Research Institute.
- **Bangalore School on Statistical Physics XIV**, ICTS & Raman Research Institute.

Teaching Experience

Teaching Assistant, IIT Mandi

- Introduction to Molecular Simulation Aug 2023 – Dec 2023
- Statistical Mechanics Jan 2024 – June 2024
- Statistical Mechanics Jan 2025 – May 2025

Professional Service

HPC Administration Team, IIT Mandi July 2021 – July 2023

- Managed user accounts and resolved user issues on the institute HPC cluster.
- Assisted users with installation of external scientific software packages.

Awards & Honours

- ANRF International Travel Support 2026 for the CECAM Flagship School "Studying dynamics in soft matter and porous materials", Germany (selected; did not attend).
- Best Poster Award, National Science Day 2024, IIT Mandi.
- Qualified CSIR-NET (Lectureship), December 2019 – 96.388 percentile.
- All India Rank 320, JEST Physics 2020 – 95.61 percentile.
- All India Rank 328 / 11,902, IIT JAM 2017 (M.Sc. admission, IIT Delhi).
- All India Rank 547 / 16,990, GATE 2020.

Certifications & Training

- Academic Writing Workshop, IIT Mandi Nov 2024
- Webinar on Science Communication, EURAXESS Oct 2023
- Space Science and Technology Awareness Training (START), ISRO Sep 2023
- MATLAB Onramp, MathWorks (verified) Feb 2022
- Scientific Writing Workshop, ACS Publications Sep 2021

References

Dr. Prasanth P. Jose (Ph.D. advisor)
Associate Professor, School of Physical Sciences, IIT Mandi
ORCID: 0000-0001-5475-2433